

02443 Stochastic Simulation

Day 1: Introduction

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Teachers

We are five teachers in this course:

Lecturer

Nicolai Siim Larsen, Assistant Professor

Regular TAs

Clara Brimnes Gardner, Postdoctoral Researcher

Jonas Bruun Hubrechts, Research Assistant

Martha Falkesgaard Nybroe, MSc Student

Support TA

Mathies Brinks Sørensen, Postdoctoral Researcher

Agenda

- 1 Structure and contents
- 2 Deliverables
- 3 Motivation
- 4 Prerequisites in probability and statistics
- 5 Group formation and brush-up sessions

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Structure

The course runs 09 - 17 every day from 8 June to 26 June and is divided into three parts:

8 June - 15 June

Lectures and exercises (Participation is mandatory)

Lectures take place in Aud. 32 and 34 in building 306

Exercise classes take place in rooms 69 - 70 - 71 - 72 in building 358

16 June - 19 June

Project 1 - Markov model of breast cancer

Group work takes place in rooms 69 - 70 - 71 - 72 in building 358

22 June - 26 June

Project 2 - Pandemic/Hospital/Queueing

Group work takes place in rooms 69 - 70 - 71 - 72 in building 358

Contents

The main pedagogical philosophy of the course is learning by doing. Consequently, the course involves a considerable amount of self-study, experimentation, and independent problem solving. The lectures provide background, motivation, and theoretical foundations, while the exercises are intended to deepen your understanding and give you hands-on experience with simulation methods and concepts.

Topics

- Random number generation
- Simulation from distributions
- Discrete event simulation
- Variance reduction methods
- Markov Chain Monte Carlo methods (MCMC methods)
- Bootstrapping methods

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Reports and deadlines

You will need to have three reports approved to pass the course:

- 1 Report with exercise solutions, deadline: Tuesday 16 June
- 2 Project 1 report, deadline: Friday 19 June
- 3 Project 2 report, deadline: Friday 26 June

All reports must be completed in groups of four students. However, each student is expected to demonstrate both individual contribution and a sound understanding of all parts of the work.

Report guidelines

Find report guidelines on DTU Learn under *Content*.

In summary:

- Concise: Include only carefully selected materials
- Document solutions: Include supporting arguments, equations, tables, and figures
- Proper presentation: Select a proper format (e.g., PDF) and write in a clear and structured manner

The main body of the exercise report should be approximately 20-30 pages depending on the number of graphs and tables. Source code should be placed in the appendix, but short code snippets and highlighted details may enter the main body of the report.

Submissions

The reports must be submitted on DTU Learn under *Assignments*.

Each submission should include 3 elements:

- 1 The report
- 2 Source code for the report
- 3 Source codes for each group member

Ideally, source code should be well organized and accompanied by appropriate comments. Most file types for source codes are acceptable e.g., R, MATLAB, and Python.

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What is stochastic simulation?

We can understand stochastic simulation in a narrow sense or a more general sense:

- Narrow sense: Generate (pseudo)random numbers with prescribed properties.
- General sense: Use a stochastic model to imitate real-world processes, systems, or events governed completely or partially by randomness.

In this course, stochastic simulation refers to running numerical experiments of various stochastic systems by generating appropriate (pseudo)random numbers.

Why study stochastic simulation?

Our reasons for studying stochastic simulation are manifold.

Stochastic simulation is a general engineering method widely applied when:

- Systems are complex
- Experiments are expensive
- System dynamics are slow

It is an excellent choice when analytical solutions cannot be obtained.

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Overview

The course requires basic knowledge of statistics, probability theory, and programming. We shall briefly review the most important concepts from probability theory:

- Basic definitions and rules
- Conditional probabilities
- Basic theorems and independence
- Random variables
- Distributions
- Moments (e.g., expected value and variance)

Basic definitions

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with

- the sample space Ω ,
- the σ -algebra (event space) $\mathcal{F}$, and
- the probability measure $\mathbb{P}$.

Then the following rules apply:

$$\mathbb{P}(\emptyset) = 0,$$

$$\mathbb{P}(\Omega) = 1,$$

and for any event $A \in \mathcal{F}$

$$0 \leq \mathbb{P}(A) \leq 1.$$

Basic rules

For any $A, B \in \mathcal{F}$ we have:

Complement rule:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Inclusion-exclusion rule for two events:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB).$$

Notation: $\mathbb{P}(AB) = \mathbb{P}(A \cap B)$.

Difference rule: If $A \subseteq B$, then

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(A).$$

Use Venn diagrams to derive these basic rules.

Conditional probabilities

We define the conditional probability of the event A given the event B by

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(AB)}{\mathbb{P}(B)}.$$

This is the probability that event A will occur given that event B has already occurred.

Multiplication rule:

$$\mathbb{P}(AB) = \mathbb{P}(A|B)\mathbb{P}(B).$$

Law of total probability (discrete case): For a finite partition of the sample space $\{B_i\}_{i=1}^n$:

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(AB_i) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

Bayes' theorem:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Independence

In probability theory, two events A and B both in $\mathcal{F}$ are independent iff.

$$\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B).$$

This is equivalent to

$$\mathbb{P}(A) = \mathbb{P}(A|B) = \mathbb{P}(A|B^c).$$

Informally, the occurrence of one event does not affect the probability of the other event occurring.

When dealing with more than two events, we distinguish (mutual/collective) independence and pairwise independence.

Random variables

We shall apply the following simplified definition of a random variable in this course:

A random variable X is a mapping $X : \Omega \rightarrow \mathbb{R}$ i.e., it is a function mapping an outcome in the sample space to a real value.

A random variable associates a possible outcome of an experiment (an $\omega \in \Omega$) with a value in the real numbers (an $x \in \mathbb{R}$).

For a (Borel) set A of $\mathbb{R}$, we then define

$$\mu(A) = \mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A)).$$

Then μ is the probability distribution of X .

In this course, the measure-theoretical foundations are not important nor required.

Discrete probability distributions

We shall work with discrete and (absolutely) continuous random variables. We can use several probability functions to describe probability distributions.

For a **discrete** random variable X

Distribution function: $F_X(x) = \mathbb{P}(X \leq x) = \sum_{y \leq x} f_X(y)$

Probability mass function: $f_X(x) = \mathbb{P}(X = x)$

For two **discrete** random variables X and Y

Joint probability mass function: $f_{XY}(x, y) = \mathbb{P}(X = x, Y = y)$

Joint distribution function: $F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \sum_{u \leq x} \sum_{v \leq y} f_{XY}(u, v)$

Marginal distribution: $f_X(x) = \mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x, Y = y)$

Conditional distribution: $f_{X|Y}(x|Y = y) = \mathbb{P}(X = x|Y = y) = \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)} = \frac{f_{XY}(x, y)}{f_Y(y)}$

Continuous probability distributions

For a **continuous** random variable X

Distribution function: $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(y) dy$

Probability density function: $f_X(x) = \frac{dF_X}{dx}(x)$

For two **continuous** random variables X and Y

Joint distribution function: $F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{XY}(u, v) dv du$

Joint probability density function: $f_{XY}(x, y) = \frac{\partial^2 F_{XY}}{\partial x \partial y}(x, y)$

Marginal distribution: $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$

Conditional distribution: $f_{X|Y}(x|Y=y) = \frac{f_{XY}(x, y)}{f_Y(y)}$

Expectation

Expectation:

$$\begin{aligned}\mathbb{E}[X] &= \int_{\Omega} X d\mathbb{P} \\ (\text{discrete case}) &= \sum_x x f_X(x) \\ (\text{continuous case}) &= \int_{-\infty}^{\infty} x f_X(x) dx\end{aligned}$$

Expectation of function:

$$\begin{aligned}\mathbb{E}[g(X)] &= \int_{\Omega} g(X) d\mathbb{P} \\ (\text{discrete case}) &= \sum_x g(x) f_X(x) \\ (\text{continuous case}) &= \int_{-\infty}^{\infty} g(x) f_X(x) dx\end{aligned}$$

Expectation is a linear operator.

Moments and variance

Moments:

$$\begin{aligned}\mathbb{E}[X^k] &= \int_{\Omega} X^k d\mathbb{P} \\ (\text{discrete case}) &= \sum_x x^k f_X(x) \\ (\text{continuous case}) &= \int_{-\infty}^{\infty} x^k f_X(x) dx\end{aligned}$$

Variance:

$$\begin{aligned}\mathbb{V}[X] &= \int_{\Omega} (X - \mathbb{E}[X])^2 d\mathbb{P} \\ (\text{discrete case}) &= \sum_x (x - \mathbb{E}[X])^2 f_X(x) \\ (\text{continuous case}) &= \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2 f_X(x) dx\end{aligned}$$

Variance can also be calculated as: $\mathbb{V}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Results on variance and covariance

Covariance:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Covariance is a bilinear operator.

Correlation:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\mathbb{V}[X]\mathbb{V}[Y]}}.$$

Variance of sum:

$$\mathbb{V}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{V}[X_i] + 2 \sum_{1 \leq j < k \leq n} \text{Cov}(X_j, X_k).$$

Independence

Two random variables are independent iff.

$$f_{XY}(x, y) = f_X(x)f_Y(y).$$

Consequently, if X and Y are independent:

- $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- $\text{Cov}(X, Y) = \text{Corr}(X, Y) = 0$
- $f_{X|Y}(x|Y = y) = f_X(x)$

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Group formation and brush-up session

The group formation session will take place here (Aud. 32 in Building 306) from 12.00 to 12.30.

The brush-up session will take place in Room 69 in Building 358 from 15.00 to 16.00 and cover various elements of hypothesis testing e.g., test statistics, critical values, p -values, and significance levels.

The brush-up session will be given by Clara Brimnes Gardner.